User Manual

Picosecond Pulsed Fiber-Coupled Laser

PLD-XXX-SM-FC-30-U



User Manual – Version v1.0.3
04 May 2026

Contents

1 General Information **2**
1.1 About This Manual . . . 2
1.1.1 User Qualification . . . 2
1.1.2 Revision History . . . 2
1.2 Symbols and Conventions . . . 2
1.3 Manufacturer Information . . . 3
1.4 Declaration of Conformity . . . 3
1.5 Product Label . . . 3
1.6 Warranty and Liability . . . 5
1.6.1 Warranty . . . 5
1.6.2 Limitation of Liability . . . 5

2 Device Description **7**
2.1 Functional Description . . . 7
2.2 Intended Use . . . 7
2.3 Reasonably Foreseeable Misuse . . . 7
2.4 Components Overview . . . 8
2.5 Technical Specifications . . . 10

3 Safety Information **11**
3.1 General Safety Instructions . . . 11
3.2 Laser Safety . . . 11
3.3 Electrical Safety . . . 12
3.4 Environmental and Operating Conditions . . . 12
3.5 Modification and Improper Use . . . 12

4 Setup **13**
4.1 Scope of Delivery . . . 13
4.2 Unpacking and Inspection . . . 13
4.3 Installation Requirements . . . 13
4.4 Mechanical Installation . . . 13
4.5 Optical Connection . . . 13
4.6 Electrical Connection . . . 14

5 Operation **15**
5.1 Principle of Operation . . . 15
5.2 Operation Modes . . . 15
5.3 Troubleshooting . . . 16

6 Maintenance and Service **17**
6.1 Cleaning . . . 17
6.2 Preventive Maintenance . . . 17
6.3 Service and Repairs . . . 17

7 Decommissioning and Disposal **18**
7.1 Decommissioning Procedure . . . 18
7.2 Disposal . . . 18

1 General Information

This chapter provides essential information on the purpose of this manual, the manufacturer, regulatory compliance, warranty conditions, and the conventions used throughout this document.

1.1 About This Manual

This manual provides instructions for the installation, configuration, operation, maintenance, and disposal of the PLD Laser Device. It forms an integral part of the product and must remain accessible to the personnel responsible for the device. If the device is transferred to a third party, all accompanying documentation must be delivered with it.

This manual contains proprietary information and may not be reproduced or distributed, in whole or in part, without prior written consent of the manufacturer.



Warning Before performing any operation, the entire manual must be read carefully and the relevant sections reviewed in detail.

1.1.1 User Qualification

The device is intended for use by qualified personnel familiar with laboratory electronic instrumentation and with laser sources and responsible for the installation, operation, and maintenance of the device throughout its service life. It is not a toy and must not be used by children under 14 years of age. Persons with reduced physical, sensory, or cognitive abilities may operate the device only under supervision by qualified personnel.

1.1.2 Revision History

Document Version: v1.0.3- 04 May 2026

A certified copy of this manual is stored in the manufacturer's technical file for the legally required period (10 years). During this period, documentation may be requested from the manufacturer.

1.2 Symbols and Conventions

This manual includes two different levels of hazard warnings. The warnings may include various safety symbols. The hazard levels and safety symbols comply with the ISO Standard for Product Safety Signs and Labels.

Hazard Levels

Warning Indicates a potential hazard that could result in serious injury or death.

Caution Indicates a potential hazard that could result in minor or moderate injury, or equipment damage.

Notice Indicates an important information that could lead to malfunction or misuse.

Safety Symbols

The following symbol types can be found on the product or in the manual.

Table 1.1: Safety symbols and their meanings

Symbol	Form	Information
	Yellow triangle with a black border and a black symbol	Indicates a hazard.
	Circular sign with a red border, diagonal line, and black symbol	Identifies a prohibited action.
	Blue circle with a white symbol	Identifies a required action.

Warning Safety pictograms performing a protective function must not be removed, covered, or damaged. If the label becomes illegible or damaged, contact authorized service for replacement.

1.3 Manufacturer Information

Manufacturer: FLIM LABS S.r.l.

Address: Via della Farnesina, 3 00135 Rome (RM), Italy

Telephone: +39-329 761540

E-mail: info@flimlabs.com

Website: www.flimlabs.com

1.4 Declaration of Conformity

The device is supplied with a Declaration of Conformity drawn up in accordance with applicable European legislation. A copy of the Declaration of Conformity is shown in Document 1.1.

1.5 Product Label

The product label affixed to the device contains the required regulatory markings and identification information including:

- Product name and model
- Manufacturer identification

EU Declaration of Conformity

CE

Device Information	
Type:	Picosecond Pulsed Laser
Model:	Picosecond Pulsed Fiber-Coupled Laser
Part Number:	PLD-405-SM-FC-30-U, PLD-445-SM-FC-30-U, PLD-488-SM-FC-30-U, PLD-520-SM-FC-30-U, PLD-635-SM-FC-30-U, PLD-808-SM-FC-30-U, PLD-850-SM-FC-30-U
Serial Number:	As indicated on the product label

Manufacturer Information	
Manufacturer:	FLIM LABS S.r.l.
Address:	Via della Farnesina, 3 00135 ROME (RM), Italy
Telephone:	+39 – 329 761540
E-mail:	info@flimlabs.com
Website:	www.flimlabs.com

The present declaration of conformity is issued under the sole responsibility of the manufacturer. The device complies with the following directives and regulations:

Applicable Directives and Regulations

- Directive 2014/35/EU, known as the “Low Voltage Directive”
- Directive 2014/30/EU, known as the “Electromagnetic Compatibility Directive”
- Directive 2011/65/EU, known as “RoHS”
- Delegated Directives (EU) 2015/863 and (EU) 2017/2102 amending Directive 2011/65/EU
- Directive 2012/19/EU, known as “WEEE”
- Regulation (EU) 2023/988 on General Product Safety

The manufacturer undertakes to provide, upon a duly reasoned request from the national authorities, all relevant information regarding the device.

Signed for and on behalf of FLIM LABS S.r.l.:

Place and Date

Rome, 24/10/2025

Signature

Alessandro Rossetta, CEO

Document 1.1: EU Declaration of Conformity

- Regulatory markings
- Electrical specifications

An example product label is shown in Figure 1.1. Disposal requirements associated with the WEEE marking are described in Section 7.2.

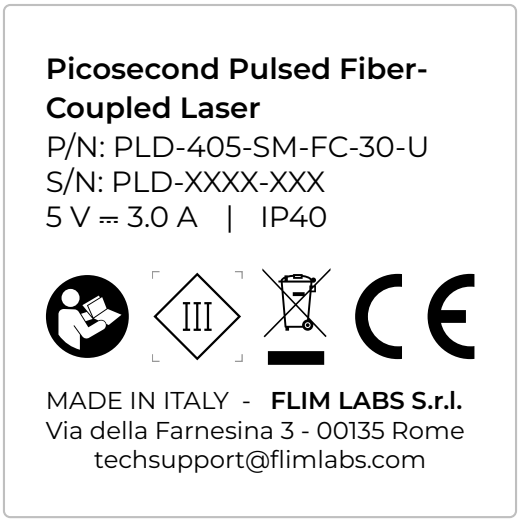


Figure 1.1: Product Label

1.6 Warranty and Liability

1.6.1 Warranty

The manufacturer provides 24 months warranty for private customers and 12 months warranty for commercial customers. The warranty period begins on the date of delivery.

The warranty covers defects in materials and workmanship under normal and intended use. It is valid only if the device is operated and maintained according to this manual. Unauthorized modification, improper use, failure to follow instructions, or repair by non-authorized personnel voids the warranty. Full warranty terms are available in the manufacturer's General Terms and Conditions published on the official website.

To submit a warranty claim, provide the following:

- Product type/version
- Proof of purchase (including date)
- Detailed description of the issue

1.6.2 Limitation of Liability

The device has been designed, manufactured, and tested in accordance with applicable European directives and internal quality standards. The manufacturer is liable exclusively for defects attributable to manufacturing faults or non-conformity with declared specifications, within the limits defined in the warranty terms.

The manufacturer shall not be liable for damages resulting from improper use, including but not limited to:

- Use outside the intended purpose
- Failure to follow this manual or applicable safety instructions
- Installation or operation by unqualified personnel
- Operation outside specified electrical or environmental limits
- Use of incompatible or non-compliant input signals or connected equipment
- Unauthorized modification, repair, or use of non-original spare parts

- Integration into systems not compliant with applicable safety regulations

The user is responsible for ensuring that the device is installed, operated, and integrated in accordance with this manual and all applicable local, national, and international regulations.

Except where otherwise required by mandatory law, the manufacturer shall not be liable for indirect, incidental, or consequential damages, including but not limited to loss of data, loss of profit, or system downtime.

2 Device Description

This section describes the functional characteristics, intended application, and technical features of the PLD Laser Device.

2.1 Functional Description

The PLD is a solid-state diode laser source that can operate in both continuous (CW) and pulsed modes with different frequencies and power settings. The laser emits at a single wavelength, which can be selected at the time of purchase from a list of available model listed in Table 2.1. The laser device is therefore available as both a visible and invisible source. The maximum laser power output allows the product to be classified as Class 3R according to the technical standard IEC EN CEI 60825-1 ed.3 (2014).

The source is a self-contained unit, designed to operate on a desktop or bench and to be connected to a microscopy instrument or other imaging equipment via optical fibers.

The source is available in different models depending on the wavelength of the laser radiation as shown in Table 2.1.

Table 2.1: Available PLD laser models.

Model	Wavelength	Radiation Type
PLD-405-SM-FC-30-U	405 nm	Visible
PLD-445-SM-FC-30-U	445 nm	Visible
PLD-488-SM-FC-30-U	488 nm	Visible
PLD-520-SM-FC-30-U	520 nm	Visible
PLD-635-SM-FC-30-U	635 nm	Visible
PLD-808-SM-FC-30-U	808 nm	Invisible
PLD-850-SM-FC-30-U	850 nm	Invisible

The model number encodes the product specifications in the following scheme:

- SM stands for single-mode
- FC for fiber-receptacle
- 30 is the laser diode (LD) power in mW
- U indicates the USB power interface

The device incorporates a laser diode capable of emitting a maximum optical power of 30 mW (class 3B), but is equipped with a control system that limits the supply current so that, even under fault conditions, the maximum laser power deliverable makes the device a class 3R laser product.

The laser emits a single-mode Gaussian beam because it is propagated inside an optical fiber with a 4 µm core.

2.2 Intended Use

The device described in this manual is designed for standalone use or incorporation into other equipment or machinery for research and diagnostic applications in the field of fluorescence analysis.

2.3 Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications

- Use with incompatible components
- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

2.4 Components Overview

Figures 2.1 and 2.2 show an overview of the key components of the device. Those include:

Front panel

1. FC/PC Output fiber connector
2. Emission status LED
   - Red: Laser emission enabled
   - Off: Laser emission disabled
3. Power Selector (%): 20; 30; 40; 50; 55; 60; 65; 70; 75; 80; 90; 100.
4. Operational Mode Selector: Off; External Synchronization; Repetition Frequency; CW.



Figure 2.1: Front panel of the Laser.

Rear panel

1. Remote Interlock
2. Power Input (USB TYPE-C)
3. Device Status LED
   - Yellow: Device powered and laser stabilization in progress
   - Green: Laser temperature stabilized and device ready for operation
   - Off: Device not powered
4. SYNC OUT 1 – synchronization external port 1
5. SYNC OUT 2 – synchronization external port 2
6. SYNC IN – synchronization internal port

SYNC Interfaces

The SYNC interfaces provide TTL-based input and output signals for triggering and monitoring the laser operation. Connections are made via SMA cables to the SYNC ports on the rear panel (see Figures 2.1 and 2.2).

SYNC IN Accepts an external TTL signal (2.5 V, 50 Ω, 50% duty cycle) when the operating mode is set to *EXT SYNC IN*. In this mode, the laser emission is triggered by the external signal source.



Figure 2.2: Back panel of the Laser.

SYNC OUT Provides a TTL output signal (2.5 V, 50 Ω, 50% duty cycle). When operating in internal frequency modes (5–80 MHz), the rising edge of this signal is synchronized with the emitted laser pulses.

When operating in *EXT SYNC IN* mode, the external trigger signal is routed to the SYNC OUT connector, allowing direct monitoring of the input signal.

2.5 Technical Specifications

Item	PLD-XXX-SM-FC-30-U
Electrical Specifications	
Electrical safety class	Class III
Power supply	USB Type-C, 5 V DC
Maximum current consumption	3 A
Maximum power consumption	15 W
Optical and Laser Specifications	
Laser type	Solid-state diode laser
Available wavelengths in nm	405 | 445 | 488 | 520 | 635 | 808 | 850
Wavelength bandwidth	±10 nm
Operating modes	CW or pulsed
Maximum output power	1.4 mW
Power stability	<1%
Pulse repetition frequency	5 MHz, 10 MHz, 20 MHz, 40 MHz, 80 MHz
Pulse duration in ps	50 | 55 | 80 | 80 | 80 | 90 | 90
Output fiber core diameter	≥ 4 µm
M^2 value	1.06 *(for 4 µm core fiber)*
Polarization	Random
Optical output	FC/PC fiber connector
Signal Specifications	
Manual command mode	Selectable repetition frequency
External synchronization input	TTL, 2.5 V, 50 Ω
Synchronization outputs	TTL, 2.5 V, 50 Ω
Physical Specifications	
Dimensions (L x W x H)	135 mm x 110 mm x 50 mm
Weight	550 g
Environmental Specifications	
Operating temperature	10°C to +40°C
Storage temperature	-20°C to +70°C
Relative humidity	80% non-condensing

Table 2.1: Technical specifications of the Picosecond Pulsed Fiber-Coupled Laser.

3 Safety Information

3.1 General Safety Instructions

Before installation and operation, ensure that this manual is complete and legible. If any part of the documentation is missing or unclear, contact the manufacturer before proceeding. The device must be used only as described in this manual. Any operation not explicitly described in this manual is considered improper use.

This manual complements applicable national and local safety regulations. If local regulations impose stricter requirements, those regulations take precedence.

Despite the implemented safety measures, residual risks may remain if the device is not used as specified.

3.2 Laser Safety

The PLD laser is a Class 3R product based on the technical standard IEC EN CEI 60825-1 ed.3 (2014), meaning a device that emits radiation that can exceed the ocular exposure limit values (MPE – Maximum Permissible Exposure) under direct viewing conditions. Therefore, this laser source may pose a risk to the eye, particularly to the retina, due to the wavelengths potentially emitted (VIS + IR-A). Damage to the retina can result in partial or total blindness.

The laser beam exits through the fiber connector, which is identified by the appropriate warning label (see Figure 3.1).

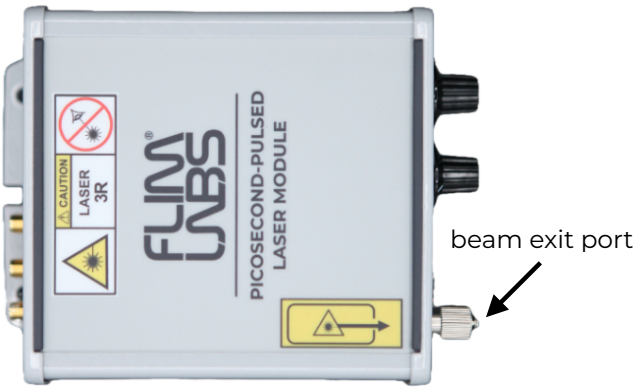


Figure 3.1: Top view of the laser showing the beam exit port.

Due care must be taken during installation and operation to avoid injury, in particular for Class 3R lasers it is prescribed to avoid direct exposure to the beam.



Warning Visible and invisible laser radiation. Avoid direct eye exposure to laser radiation.
Do not turn on the laser source if the fiber optic output connector is not correctly connected with a closed optical path.
Do not point the laser output directly at people.
Do not open the laser source.
Use the laser source only under safe conditions as outlined in Sections 3.3 and 3.4 .
Pay attention to the warning labels on the laser source.

Laser Classification Labels

The classification labels attached to the product are shown in Table 3.1:

Table 3.1: Laser classification labels

Label	Classification
	3R classification label for visible radiation
	3R classification label for invisible radiation
	Alternative classification label for Class 3R laser

3.3 Electrical Safety

- Operate the device only with the specified power supply.
- Use undamaged, properly rated cables.
- Disconnect power before modifying signal connections.
- Do not open the enclosure. There are no user-serviceable internal components.
- Keep away from water sources and other liquids.
- Unauthorized repair or modification is not permitted.

3.4 Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 10°Cto +40°C
- Storage temperature: −20°Cto +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases. Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.

3.5 Modification and Improper Use

The device must not be modified, altered, or repaired by unauthorized personnel. Use outside the intended application, operation beyond specified electrical limits, or failure to follow the procedures described in this manual may result in malfunction and void the warranty.

4 Setup

This section describes the correct procedure for unpacking, installing, and configuring the device before operation.

4.1 Scope of Delivery

- PLD module
- SMA coaxial cable (fixed delay line)
- Optical Fiber (single mode)
- USB Type-C power cable
- User manual
- Declaration of Conformity

Verify that all items are present before proceeding with installation.

4.2 Unpacking and Inspection

The device is supplied in protective packaging to protect it during storage and transport. Handle the device carefully during unpacking to avoid mechanical damage.

After removing the packaging:

- Inspect the device and accessories for visible damage.
- Check for dents, deformation, cracks, or signs of impact.
- Verify that connectors and cables are intact.

The device is compact and lightweight and can be moved manually. During handling, support it securely to prevent accidental dropping. If any damage, missing parts, or abnormalities are detected, stop the installation and contact the manufacturer before proceeding.



Warning Keep packaging materials out of reach of children and dispose of them according to local regulations.

4.3 Installation Requirements

Before installation and operation, ensure that the environmental conditions specified in Section 3.4 are met. Children must not have access to the installation or storage area.

The device requires a regulated 5 V DC power source via USB Type-C capable of supplying at least 3 A.

4.4 Mechanical Installation

Place the device on a stable laboratory surface with adequate space for cable routing.

Ensure that:

- The device is positioned to prevent mechanical stress on connectors.
- Cables are routed safely and do not create tripping hazards.
- The power supply location is dry and away from water sources (e.g., sinks or splash zones).

Remove any protective dust or moisture covers before operation.

4.5 Optical Connection

Connect one end of the fiber optic cable to the FC/PC fiber OUT port of the device (see Figure 2.1) and connect the other end to the optical system. Ensure proper connector mating and avoid mechanical stress on the fiber.

4.6 Electrical Connection

All electrical connections must be performed by qualified personnel who have read this manual. The connector locations are shown in Figure 2.2.



Warning Before connecting the USB Type-C cable to the device's power input, make sure the operating mode switch is set to OFF.
Ensure that the fiber optic cable is connected to the beam output port.



Notice Before connecting the device, verify that the input signal characteristics are compatible with the technical specifications as listed in Table 2.1. Improper signal conditions may result in malfunction or unreliable operation.

Proceed as follows:

1. Power Connection:
   a) Before connecting the USB Type-C cable to the power input of the device, make sure that the operating mode selector is set to OFF;
   b) Connect the USB Type-C cable to the device power input.
   c) Connect the other end to a certified 5 V DC USB power adapter.
   d) Verify that the Power LED is illuminated.
2. Connect the SMA cable to the *SYNC IN* port, to control the laser pulse with an external signal.
3. Connect the SMA cable to one of the *SYNC OUT 1* or *SSYNC OUT 2* ports, to monitor the laser pulse or to synchronize any acquisition electronics.

5 Operation

This section describes the functional behavior of the PLD Laser during use, including its operating principle, normal operating conditions, adjustable parameters, and troubleshooting guidance.

The device does not require initial configuration. Once the mechanical, electrical, and optical installation has been completed, as indicated in Chapter 4, the device is ready for operation.

Warning Do not operate the device if it is damaged or shows signs of unsafe condition.

5.1 Principle of Operation

The PLD device is a laser source that can operate in both continuous and pulsed modes. The operating mode and the output power can be configured using the operating mode selector and the laser power selector (see Figure 2.1).

5.2 Operation Modes

The following modes can be selected:

OFF

If the operating mode selector is set to OFF, the laser does not emit light.

Continuous Mode (CW)

- Set the operating mode selector to CW;
- Set the power selector to the desired power percentage.

Pulsed Mode

In pulsed operation, the laser repetition rate can be set manually or using an external trigger signal.

1. Manual Setting
   - Set the operating mode selector to the desired frequency: 5, 10, 20, 40, 80 MHz;
   - Set the power selector to the desired power percentage.
2. External Trigger
   - Set the operating mode selector to EXT SYNC IN (see Figure 2.1)
   - Connect an external TTL trigger signal via the SYNC IN port on the rear panel (Figure 2.2)
   - Set the power selector to the desired power percentage.

The SYNC OUT 1 and/or SYNC OUT 2 ports on the rear panel can be used to monitor the laser pulse or to synchronize acquisition electronics.

Notice Laser stability and pulse shape may be reduced during initial warm-up of the device. Wait for ten minutes after starting laser emission to let the system stabilize before performing sensitive measurements.

5.3 Troubleshooting

Common faults and corrective actions are summarized in Table 5.1.

Table 5.1: Overview of common problems and recommended corrective actions

Problem	Possible Cause	Recommended Action
Power and Device Status		
Device does not power on	Insufficient USB power	Use a USB port capable of delivering at least 900 mA at 5 V DC.
Device does not power on	Defective USB cable	Replace the USB cable and verify proper connector seating.
Input Signal		
No laser radiation output	Input signal amplitude too low (SYNC IN)	Verify that the external TTL signal meets the required amplitude, impedance, and duty-cycle specifications.
No laser radiation output	Improper impedance matching	Use 50 Ω coaxial cables and ensure proper termination.
Output Signal		
No SYNC OUT signal detected by acquisition system	Incorrect output connection	Verify that the SMA output is connected to the correct input channel of the acquisition system.
Output signal distorted	Incorrect cabling or excessive cable length	Use short 50 Ω coaxial cables and avoid adapters or T-connectors.



Warning If any unexpected or potentially hazardous anomaly occurs during startup or operation, switch off the device immediately and disconnect it from the power source. Contact technical support before further use.

6 Maintenance and Service

This section describes the cleaning, preventive maintenance, corrective maintenance, and authorized service procedures for the device. All maintenance activities must be performed by qualified personnel who have read and understood this manual.



Warning Before performing any cleaning or physical maintenance operation, the device must be switched off and disconnected from the power source.

Maintenance must be carried out in compliance with this manual and applicable workplace safety regulations. The maintenance area should provide adequate illumination; a minimum lighting level of 200 lux is recommended.

6.1 Cleaning

The device requires periodic external cleaning to ensure reliable operation.

- Remove dust or dirt from the exterior surfaces using a dry or slightly damp lint-free cloth.
- A mild, neutral detergent may be used if necessary.
- Do not allow liquids to enter connectors or internal components.
- Avoid the use of solvents, aggressive chemicals, or abrasive materials.

For connectors and hard-to-reach areas (e.g., USB ports), a soft-bristled brush or compressed air may be used.

6.2 Preventive Maintenance

The device does not require routine internal maintenance.
Periodically inspect:

- External surfaces for mechanical damage
- Connectors and cables for wear or deformation
- Product label for legibility. See Section 1.5 for label content and handling requirements.

If any abnormal condition is detected, discontinue use and contact technical support.

6.3 Service and Repairs

The device contains no user-serviceable internal components. In the event of malfunction, accidental damage, or improper operation:

- Switch off and disconnect the device.
- Do not attempt internal repair.
- Contact the manufacturer's technical service for instructions.

Repairs or modifications must be carried out exclusively by FLIM LABS S.r.l. or authorized personnel. Use only original FLIM LABS spare parts. Further warranty conditions are described in Section 1.6.1.



Warning Unauthorized intervention or the use of non-original components may compromise safety, performance, and warranty coverage.



Warning Opening the case may result in laser exposure.

7 Decommissioning and Disposal

This section describes procedures for removing the device from service and disposing of it in accordance with applicable regulations.

7.1 Decommissioning Procedure

The device is designed for long-term laboratory use. When it reaches the end of its technical or operational life, it must be decommissioned to prevent further use.

Decommissioning requires:

- Disconnecting the device from all power sources.
- Removing all signal connections.
- Ensuring that the device cannot be unintentionally reused.

The same procedure applies in cases of:

- Permanent withdrawal from service
- Storage without intended future use
- Final disposal

Reuse of components after final decommissioning is not permitted unless explicitly authorized by the manufacturer.

7.2 Disposal

The device must be disposed of in accordance with applicable local and European regulations governing electrical and electronic equipment. In accordance with Directive 2012/19/EU (WEEE), the device must not be disposed of with household waste. The product label includes the WEEE symbol indicating the requirement for separate collection of electrical and electronic equipment (see Section 1.5). Contact your local waste management authority or certified collection center for proper disposal procedures.

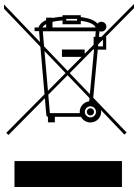


Figure 7.1: WEEE symbol